

Fig. 2: Actual-size PCB layout for the counter

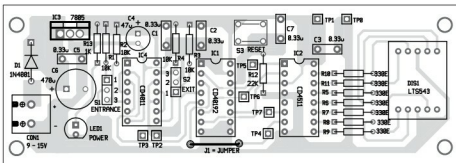


Fig. 3: Component layout for the PCB

before use.

Resistors R5 through R11 are usually in the range of 120 to 470 ohms, depending on the type of DIS1. Here,

we have used 330-ohm resistors.

Power supply of the circuit is built around linear regulator 7805 (IC3). The input voltage is applied to connector

CON1 and can be AC or DC in the range 9-15V.

## Construction and testing

An actual-size, single-side PCB for the difference counter is shown in Fig. 2 and its component layout in Fig. 3.

Mount the components on the PCB to save time and avoid assembly errors. Carefully assemble the components and double check for any overlooked error. Check/verify the voltages at various points as per the table.

You may use any type of SPDT switches (S1 and S2) to check the increment and decrement in DIS1 display. But for practical application, S1 and S2 should be SPDT foot switches, which could be fitted in level with the floor at the entrance and exit doors, respectively. ●

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